AQ2.3: Activity Questions 3 - Not Graded

This assignment will not be graded and is only for practice.
Level 1:

1 point
Match the matrices in Column A with their inverses in Column B.

- [ ] a $\rightarrow$ → iii
- [ ] a $\rightarrow$ → ii
- [ ] b $\rightarrow$ → iii
- [ ] b $\rightarrow$ → i
- [ ] c $\rightarrow$ → i
- [ ] c $\rightarrow$ → ii

No, the answer is incorrect.
Score: 0
Accepted Answers:
a $\rightarrow$ → ii
b $\rightarrow$ → iii
c $\rightarrow$ → i

(Use the below information to answer questions 2 and 3).
Consider a system of linear equations

$$\begin{aligned} 2x_1 - x_2 &= 3 \\ x_1 - x_3 &= 3 \\ x_2 - x_3 &= 2 \end{aligned}$$

Let the matrix representation of the above system be $Ax = b$, where $A = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, and $b = \begin{bmatrix} 3 \\ 3 \\ 2 \end{bmatrix}$.

1 point
Choose the set of correct options.

- [ ] $A^{-1} = \begin{bmatrix} 2 & -1 & 0 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$
- [ ]

A^{-1} = [1−11 1−22 1−21]A−1 = 111−1−2−2121

☐

Adjoint of the matrix AA is [1−11 1−22 1−21] 111−1−2−2121

☐

$det(A)$ = 1det(A) = 1.

No, the answer is incorrect.
Score: 0
Accepted Answers:

A^{-1} = [1−11 1−22 1−21]A−1 = 111−1−2−2121

Adjoint of the matrix AA is [1−11 1−22 1−21] 111−1−2−2121

$det(A)$ = 1det(A) = 1.

1 point

Choose the set of correct options.

☐

$x_1 = -2$x1 = −2.

☐

$x_2 = 1$x2 = 1.

☐

$x_3 = -1$x3 = −1.

☐ None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$x_2 = 1$x2 = 1.

$x_3 = -1$x3 = −1.

1 point

Choose the set of correct options.

☐

A system of linear equations $Ax = b$Ax = b is called a homogeneous system of linear equations if $b \neq 0$b = 0.

☐

A system of linear equations $Ax = b$Ax = b is called a non-homogeneous system of linear equations if $b \neq 0$b = 0

☐

If vv is a solution of the system of linear equations $Ax = b$Ax = b, then $\frac{1}{c}v$c1v is a solution of system of linear equations $cAx = b$cAx = b, where $c \neq 0$c = 0.

☐

Let $Ax = b$ be a system of linear equations. If A is invertible, then $adj(A)x = b$ also has a solution.

No, the answer is incorrect.
Score: 0
Accepted Answers:

A system of linear equations $Ax = b$ is called a non-homogeneous system of linear equations if $b \neq 0$

If v is a solution of the system of linear equations $Ax = b$, then $\frac{1}{c}v$ is a solution of system of linear equations $cAx = b$, where $c \neq 0$.

Let $Ax = b$ be a system of linear equations. If A is invertible, then $adj(A)x = b$ also has a solution.

Suppose C is the cofactor matrix of $\begin{bmatrix} 3 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$. Let C_{ij} be the ij-th element of the cofactor matrix C. Answer the questions 5,6 and 7.

Find the value of C_{12}.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1

1 point

Find the value of C_{22}.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) -3

1 point

Find the sum of the elements in the third row of C.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0

1 point

Level 2:

1 point

Which of the following square matrices of order 3 are the same as their adjoint matrices?

- ☐ Identity matrix.
- ☐ Zero matrix.
- ☐ Any scalar matrix.

- [ ] Any diagonal matrix.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Identity matrix.
Zero matrix.

1 point
Choose the set of correct options.

[Hint: For the first option, find out the adjoint matrix of an arbitrary square upper triangular matrix of order 2, i.e., take the matrix $\begin{bmatrix} a & b \\ 0 & d \end{bmatrix}$ and find out its adjoint matrix.]

- [ ] The adjoint of a 2×2 real upper triangular matrix is an upper triangular matrix.
- [ ] There exists a square matrix of order 3, such that $A = A^{-1} = adj(A)$.
- [ ] If $A = A^{-1}$, then $det(A)$ must be 1.
- [ ] If $A = A^{-1}$, then A must be an identity matrix.
- [ ] If $A = A^{-1}$, then $A^2 = I$.
- [ ] If $A^{-1} = adj(A)$, then $det(A)$ must be 1.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The adjoint of a 2×2 real upper triangular matrix is an upper triangular matrix.
There exists a square matrix of order 3, such that $A = A^{-1} = adj(A)$.
If $A = A^{-1}$, then $A^2 = I$.
If $A^{-1} = adj(A)$, then $det(A)$ must be 1.

1 point
If A and B are squares matrices of order 2, then choose the set of correct options.
[Hint: Find out the adjoint matrix of an arbitrary square matrix of order 2, i.e., take the matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and find out its adjoint matrix.]

- [ ] $adj(AB) = adj(A)adj(B)$.
- [ ] $adj(AB) = adj(B)adj(A)$.
- [ ] $adj(A + B) = adj(A) + adj(B)$.
- [ ] $adj(A^T) = adj(A)^T$.
- [ ] $adj(A^{-1}) = adj(A)^{-1}$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$adj(AB) = adj(B)adj(A)$.
$adj(A + B) = adj(A) + adj(B)$.

$adj(A^T) = adj(A)^T$ adj(AT) = adj(A)T .
$adj(A^{-1}) = adj(A)^{-1}$ adj(A−1) = adj(A)−1.

Check Answers
Your score is: 0/10